

Editors-in-Chief

*Biomechanics and Modeling in Mechanobiology*

Dear Editors,

Please consider our manuscript, “Comparing deformation subspaces across anatomical variability: a population finite-element study of the human pelvis,” for publication as an Original Article in *Biomechanics and Modeling in Mechanobiology*.

Population finite-element studies require comparable mechanical descriptors across anatomically different subjects. Eigenvalue rank is convenient, but neighbouring modes can exchange their correspondence to anatomical deformation patterns. A rank-based comparison may consequently suggest a large change in anatomical coupling even when the relevant modal subspace retains similar deformation capacity.

We investigate this problem in 278 CT-derived pelvic finite-element models using a common reference domain, spectral gap analysis, modal assignment, principal angles, basis-invariant anatomical functionals, and static modal energy decomposition. Low-rank modes retain stable ordering, whereas a recurrent rank 9–10 block occurs in 54.7% of subjects. Within this block, median anteroposterior inlet coupling at rank 9 differs by approximately 69% between subjects without and with label exchange; the corresponding subspace capacity differs by approximately 5%. Standing and internal-ring expansion loads recruit different spectral regions, connecting the comparison framework to controlled pelvic loading.

Building on earlier pelvic spectral metrics and established subspace methods, the contribution is a quantitative demonstration of how mode correspondence affects anatomical interpretation across a pelvic population. We believe this fits the journal’s interest in computational solid mechanics and mathematical methods that improve biomechanical analysis. The study builds on established subject-specific modelling and addresses an interpretive problem relevant to readers working with anatomical variability and reduced representations of musculoskeletal mechanics.

We distinguish kinematic capacity from force-dependent compliance and retain the limitations of the linearised formulation. The expansion loads are comparative mechanical probes, not simulations of childbirth. A constructed analytical example illustrates the spectral mechanism. The main article presents the mathematical formulation, subspace comparisons, anatomical capacity optimisation, uncertainty propagation, and analytical model. All figures, tables, extended results, and validation checks are included in the manuscript. No supplementary files accompany the article.

The source anatomical dataset is cited separately from the present analysis. Summary results are supplied within the manuscript; a public archive of the custom code and additional derived outputs has not yet been established, as stated in the availability declarations.

Thank you for considering our manuscript.

Sincerely,

Niels Hammer, on behalf of the authors

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